Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

In a double-slit experiment the angular width of a fringe is found to be 0.2° on a screen $1\ \mathrm{m}$ away. The wavelength of light used is $6000\mathring{A}$. What will be the angular width of a fringe if the entire experimental arrangement is immersed in water? Refractive index of water $= \frac{4}{3}$.

(A) 0.15°

(B) 0.18°

(C) 0.2°

(D) 0.27°

Q.2

The condition for obtaining secondary maxima in the diffraction pattern due to single slit is

(A) $\mathrm{a}\sin\theta = \mathrm{n}\lambda$

(B) $a\sin\theta = (2n-1)\frac{\lambda}{2}$

(C) $\mathrm{a}\sin\theta = (2\mathrm{n}-1)\lambda$

(D) $\mathrm{a}\sin\theta = \frac{\mathrm{n}\lambda}{2}$

Q.3

In a single slit diffraction experiment, the width of the slit is made double its original width. Then the central maxima of the diffraction pattern will become

(A) narrower and fainter

(B) narrower and brighter

(C) broader and fainter

(D) broader and brighter

Q.4

At the first minimum adjacent to the central maximum of a single-slit diffraction pattern, the phase difference between the Huygen's wavelet from the edge of the slit and the wavelet from the midpoint of the slit is :

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(A) $\frac{\pi}{2}$ radian

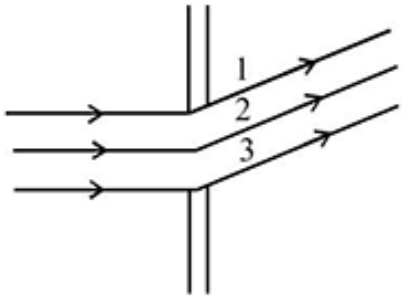
(B) π radian

(C) $\frac{\pi}{8}$ radian

(D) $\frac{\pi}{4}$ radian

Q.5

The adjacent figure shows Fraunhoffer's diffraction due to a single slit. If first minimum is obtained in the direction shown, then the path difference between rays 1 and 3 is



(A) 0

(B) $\lambda/4$

(C) $\lambda/2$

(D) λ

Q.6

The box of a pin hole camera, of length L, has a hole of radius a. It is assumed that when the hole is illuminated by a parallel beam of light of wavelength λ the spread of the spot (obtained on the opposite wall of the camera) is the sum of its geometrical spread and the spread due to diffraction. The spot would then have its minimum size (say $\mathrm{b_{min}}$) when :

(A) $\mathrm{a} = \sqrt{\lambda \mathrm{L}}$ and $\mathrm{b_{min}} = \sqrt{4\lambda \mathrm{L}}$

(B) $\mathrm{a} = \frac{\lambda^2}{\mathrm{L}}$ and $\mathrm{b_{min}} = \sqrt{4\lambda \mathrm{L}}$

(C) $\mathrm{a} = \frac{\lambda^2}{\mathrm{L}}$ and $\mathrm{b_{min}} = \left(\frac{2\lambda^2}{\mathrm{L}}\right)$

(D) $\mathrm{a} = \sqrt{\lambda}\,\mathrm{l}$ and $\mathrm{b_{min}} = \left(\frac{2\lambda^2}{\mathrm{L}}\right)$

Q.7

In Fresnel's biprism experiment the width of fringes is $2\ \mathrm{cm}$ which are formed at a distance of 2 meter from the slit. If the wavelength of light is $5100\mathring{A}$ then the distance between two coherent sources will be

(A) $5.1 \times 10^{-4}\ \mathrm{m}$

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(B) 5.1×10^{4} cm

(C) 5.1×10^{-4} mm

(D) 10.1×10^{-4} cm

Q.8

A physics professor wants to find the diameter of a human hair by placing it between two flat glass plates, illuminating the plates with light of vaccum wavelength $\lambda = 552$ nm and counting the number of bright fringes produced along the plates. The Professor find 125 bright fringes between the edge of the plates and the hair. What is the diameter of the hair?

(A) 525×10^{-9} m

(B) 344×10^{-3} m

(C) 3.44×10^{-5} m

(D) None of the above

Q.9

The width of the diffraction band varies

(A) inversely as the wavelength

(B) directly as the width of the slit

(C) directly as the distance between the slit and the screen

(D) inversely as the size of the source from which the slit is illuminated

Q.10

The first diffraction minimum due to the single slit diffraction is seen at $\theta = 30^{\circ}$ for a light of wavelength $5000\mathring{A}$ falling perpendicularly on the slit. The width of the slit is

(A) 2.5×10^{-5} cm

(B) 1.25×10^{-5} cm

(C) 10×10^{-5} cm

(D) 5×10^{-5} cm

Q.11

The angle substanded by the first diffraction minimum for a point source viewed in the hydrogen line at 1420MHz with a radio telescope having an aperture of 25 m is :

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(A) 0.8°

(B) 0.64°

(C) 1.2°

(D) 2.2°

Q.12

When the diffraction pattern from a certain slit illuminated with laser light $(\lambda = 6330\ \mathrm{A}^\circ)$ is projected on a screen $150\ \mathrm{cm}$ from the slit, the second minima on each side are separated by $8\ \mathrm{cm}$. This tells us that:

(A) the slit is approximately $0.005\ \mathrm{cm}$ wide

(B) the slit is approximately $0.05\ \mathrm{cm}$ wide

(C) a $/\lambda$ is approximately 7.5 (a is the slit width)

(D) a/λ is approximately 750

Q.13

In an experiment of single slit diffraction pattern, first minimum for red light coincides with first maximum of some other wavelength. If wavelength of red light is $6600\mathring{A}$, then wavelength of first maximum will be:

(A) $3300\mathring{A}$

(B) $4400\mathring{A}$

(C) $5500\mathring{A}$

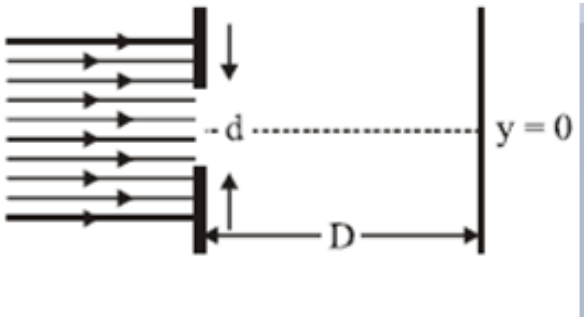
(D) $6600\mathring{A}$

Q.14

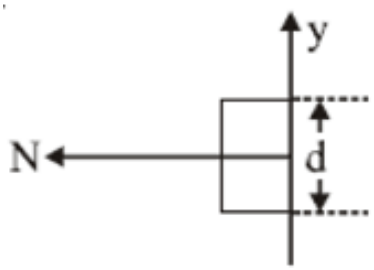
In an experiment, electrons are made to pass through a narrow slit of width d comparable to their de Broglie wavelength. They are detected on a screen at a distance D from the slit (see figure). Which of the following graphs can be expected to represent the number of electrons N detected as a function of the detector position $y(y = 0$ corresponds to the

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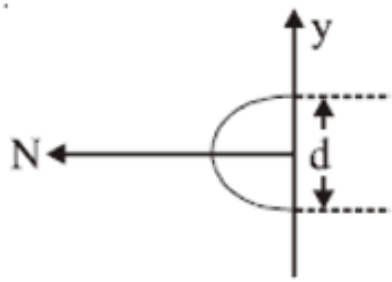
middle of the slit) ?



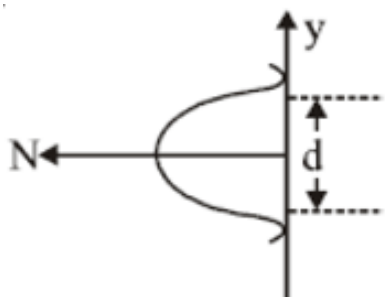
(A)



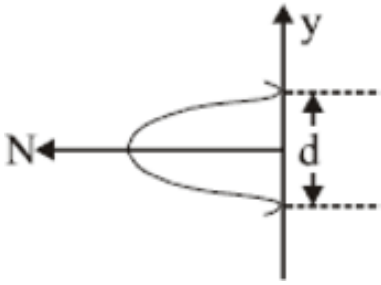
(B)



(C)



(D)



Q.15

In Young's double slit experiment, the two slits act as coherent sources of equal amplitude A and wavelength λ. In another experiment with the same set up the two slits are sources of equal amplitude A and wavelength λ but are incoherent. The ratio of the intensity of light at the mid-point of the screen in the first case to that in the second case is :

(A) $4:1$

(B) $2:1$

(C) $1:1$

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(D) None of the above

Answers & Solutions

Q.1 Answer:

0.15°

Solution:

The angular width of a fringe is given by

$\theta = \frac{\lambda}{d}$

In air:

$\theta_a = \frac{\lambda_a}{d}$

In water: $\theta_w = \frac{\lambda_w}{d}$

$\therefore \quad \frac{\theta_w}{\theta_a} = \frac{\lambda_w}{\lambda_a}$ **or** $\theta_w = \frac{\lambda_w \theta_a}{\lambda_a}$ **(1)**

Now, refractive index is defined as

$$\mu_w = \frac{\text{speed of light in air}}{\text{speed of light in water}}$$

$$= \frac{v_a}{v_w} = \frac{v\lambda_a}{v\lambda_w} = \frac{\lambda_a}{\lambda_w} \quad (2)$$

where v is the frequency of light which remains unchanged. Using Eq. (2) in Eq. (1) we have

$\theta_w = \frac{\theta_a}{\mu_w} = \frac{0.2^\circ}{4/3} = 0.15^\circ$

So the correct choice is (a).

Q.2 Answer:

$a \sin\theta = (2n-1)\frac{\lambda}{2}$

Solution:

Correct option is (b) a $\sin\theta = (2n-1)\frac{\lambda}{2}$

Condition for nth order maxima is a $\sin\theta = \left(\mathrm{n} + \frac{1}{2}\right)\lambda, \mathrm{n} = \mathrm{o}, 1, 2 \ldots$

If n starts from 1, then the condition is a $\sin\theta = (2n-1)\frac{\lambda}{2}$

Q.3 Answer:

narrower and brighter

Solution:

The width of the central maximum is given by

$\beta = \frac{2\lambda \mathrm{D}}{\mathrm{d}}$

$\Rightarrow$ **If** $\mathrm{d} \rightarrow 2\,\mathrm{d}$, **then** β **decreases.**

Also, intensity $\mathrm{I} = \mathrm{I}_0 \left[\frac{\sin\alpha}{\alpha}\right]^2$

where $\alpha = \frac{\pi \mathrm{d} \sin\theta}{\lambda}$

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∴ **I increases as d increases**

∴ **The central maximum will become narrower and brighter.**

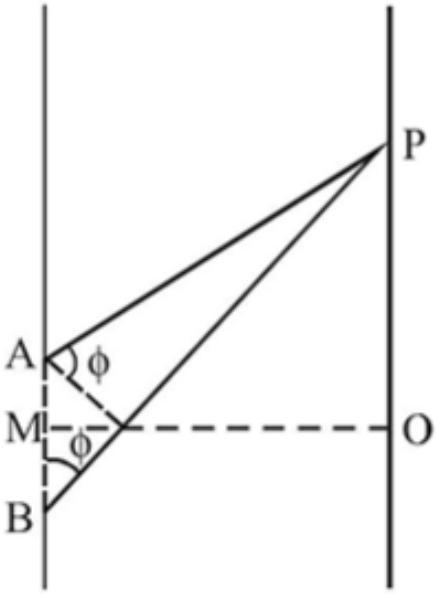
Q.4 Answer:

π radian

Solution:

For first minima at P $\mathrm{AP} - \mathrm{BP} = \lambda$

$\mathrm{AP} - \mathrm{MP} = \frac{\lambda}{2}$



So phase difference, $\phi = \frac{2\pi}{\lambda} \times \frac{\lambda}{2} = \pi$ **radian**

Q.5 Answer:

$\lambda/2$

Solution:

In Fraunhoffer diffraction, for minimum intensity,

$\Delta \mathrm{x} = \mathrm{m}\frac{\lambda}{2}$

For first minimum, $\mathrm{m} = 1$

$\therefore \Delta \mathrm{x} = \frac{\lambda}{2}$

Q.6 Answer:

$\mathrm{a} = \sqrt{\lambda \mathrm{L}}$ and $\mathrm{b}_{\min} = \sqrt{4\lambda \mathrm{L}}$

Solution:

Given geometrical spread = a

Diffraction spread $= \frac{\lambda}{\mathrm{a}} \times \mathrm{L} = \frac{\lambda \mathrm{L}}{\mathrm{a}}$

The sum $\mathrm{b} = \mathrm{a} + \frac{\lambda \mathrm{L}}{\mathrm{a}}$

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For b to be minimum $\frac{\mathrm{db}}{\mathrm{da}}=0 \quad \frac{\mathrm{d}}{\mathrm{da}}\left(\mathrm{a}+\frac{\lambda \mathrm{L}}{\mathrm{a}}\right)=0$

$$\mathrm{a}=\sqrt{\lambda \mathrm{L}}$$

$$\mathrm{b\,min}=\sqrt{\lambda \mathrm{L}}+\sqrt{\lambda \mathrm{L}}=2\sqrt{\lambda \mathrm{L}}=\sqrt{4\lambda \mathrm{L}}$$

Q.7 Answer:

$5.1 \times 10^{-4}\ \mathrm{m}$

Solution:

$d=\frac{D\lambda}{\beta}$

According to question,

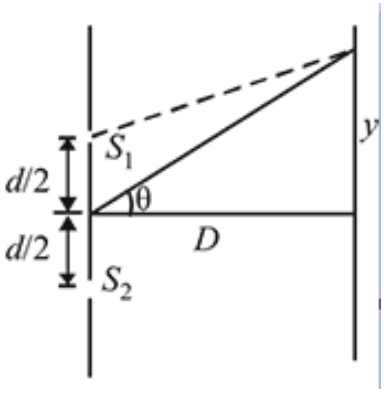
$\lambda=5100 \times 10^{-10}\ \mathrm{m}$

$\beta=\frac{2}{10} \times 10^{-2}\ \mathrm{m} \quad \ldots$ **(ii)**

$D=2\ \mathrm{m}, d=?$

From eqs. (i) and (ii)

$d=\frac{2\times 51\times 10^{-8}}{2\times 10^{-3}}=5.1 \times 10^{-4}\ \mathrm{m}$



Q.8 Answer:

$3.44 \times 10^{-5}\ \mathrm{m}$

Solution:

The reflections from the boundaries will cause a net 180° **phase shift. The condition for bright fringes is** $2t=(\mathrm{m}+1/2)\lambda_{\mathrm{film}}$

Now, $\mathrm{m}=124$ **since there is a bright fringe for** $\mathrm{m}=0$ **and** $\lambda_{\mathrm{film}}=\frac{\lambda}{\mathrm{n}}$

$t=\frac{\left(m+\frac{1}{2}\right)\lambda_{film}}{2}=\frac{\left(m+\frac{1}{2}\right)\lambda}{2n}$

$=\frac{\left(124+\frac{1}{2}\right)\left(552\times 10^{-9}m\right)}{2\times(1.00)}=3.44 \times 10^{-5}m$

Q.9 Answer:

directly as the distance between the slit and the screen

Solution:

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The width of the diffraction band is given by $\beta = \frac{\lambda D}{d} \Rightarrow \beta \propto \lambda$
$\beta \propto \mathrm{D}$ **and** $\beta \propto \frac{1}{d}$

Q.10 Answer:

10×10^{-5} cm

Solution:

For first minimum, $d \sin\theta = \lambda$
$\Rightarrow \quad d = \frac{\lambda}{\sin\theta} = \frac{5000 \times 10^{-8} \text{ cm}}{\sin 30^\circ}$
$= \frac{5000 \times 10^{-8} \text{ cm}}{1/2} = 10 \times 10^{-5}$ cm

Q.11 Answer:

1.2°

Solution:

$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{1420 \times 10^6} = 0.214$ m
$\sin\theta = \frac{1.22\lambda}{d} = \frac{1.22 \times 0.219}{25} = 0.010$
or $\theta = 0.6^\circ$**, and** $2\theta = 1.2^\circ$

Q.12 Answer:

the slit is approximately 0.005 cm wide

Solution:

$\sin\theta_2 = \frac{2\lambda}{d}$ **or** $\theta_2 = \sin^{-1}\left(\frac{2\lambda}{d}\right)$
Given $y = \mathrm{D}(2\theta_2) = 8 \times 10^{-2}$
or $1.5 \times 2 \times \sin^{-1}\left(\frac{2\lambda}{d}\right) = 8 \times 10^{-2}$
$\Rightarrow d \simeq 0.005$ cm

Q.13 Answer:

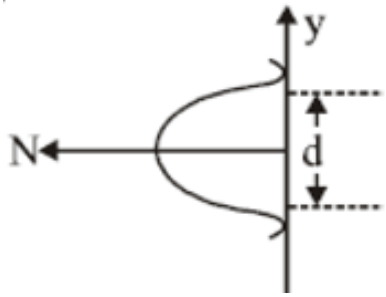
$4400\mathring{A}$

Solution:

In a single slit experiment, For diffraction maxima, $a \sin\theta = (2\mathrm{n}+1)\frac{\lambda}{2}$ **and for diffraction minima,** $a \sin\theta = \mathrm{n}\lambda$ **According to question,** $(2 \times 1 + 1)\frac{\lambda}{2} = 1 \times 6600 \quad \left(\because \lambda_\mathrm{R} = 6600\mathring{A}\right)$
$\lambda = \frac{6600 \times 2}{3} = 4400\mathring{A}$

Q.14 Answer:

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Solution:

Width of central maxima $= \left(\frac{2\lambda D}{d}\right) \because \quad \lambda \approx d$

$\Rightarrow$ **width of central maxima will be more than** d

Q.15 Answer:

2: 1

Solution:

$$\frac{I_{\text{coherent}}}{I_{\text{noncoherent}}} = \frac{4I_0}{2I_0} = \frac{2}{1}$$

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